

ARJUNA × NMIT

SAPTA KRISHI MODULAR GRAND CHALLENGE

BHARAT AGRI-ROBOTICS MODULAR CHALLENGE 2026-27

Design. Swap. Deploy. Feed Bharat.

Seven agricultural robot missions — plant health, field transport, weeding, seeding, arecanut climbing, coconut climbing and river-weed cleaning — built on one modular robotics system: the ARJUNA Autonomous Mobile Robot and the Jetrick reconfigurable robot controller.

Organised by	Nitte Meenakshi Institute of Technology (NMIT), Nitte (Deemed to be University), Bengaluru
Kit 1	ARJUNA AMR — fully assembled · 2D LiDAR · OAK-D 3D stereo depth camera · NVIDIA Jetson Orin Nano Super 8 GB
Kit 2	Jetrick Robot Controller — reconfigurable; teams design their own robot body for the task
Proposed partners	[Platform partner(s) — TBC] · [Agricultural university — TBC] · [Agri-tech industry — TBC] · [FPO / plantation partner — TBC]
Academic partners	Department of Robotics & Artificial Intelligence · Centre for Robotics Research · Co-Innovation Centre / DeepTech Innovation Ecosystem
Proposed cycle	October 2026 – April 2027

Version 1.0 — Draft for Partner Review

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1. EXECUTIVE SUMMARY

Indian agriculture does not have one robotics problem — it has many. A farmer needs help sowing, weeding, carrying, checking plant health and harvesting; plantation owners struggle to find people willing to climb arecanut and coconut trees; and invasive water weeds choke the lakes and canals that feed irrigation. Building a separate expensive machine for each job is not affordable for most Indian farms.

The **ARJUNA Sapta Krishi Modular Grand Challenge** ('Sapta' = seven) asks a different question:

Can one modular robotics system — a common intelligent brain and interchangeable bodies and tools — solve seven real Indian agricultural problems, affordably and autonomously?

Teams compete using two kits:

- **Kit 1 — ARJUNA Autonomous Mobile Robot:** a fully assembled robot with 2D LiDAR, an OAK-D 3D stereo depth camera and an NVIDIA Jetson Orin Nano Super (8 GB). Hardware is fixed, so the competition is about intelligence.
- **Kit 2 — Jetrick Robot Controller:** a reconfigurable controller that lets teams design and build their own robot body — a tree climber, a floating weed collector, a weeder or a seeder.

Both kits connect through a common **Krishi Module Interface (KMI)**, so a tool built on Jetrick can be mounted on ARJUNA and recognised automatically. The challenge ends with the **Krishi Mela Farm Day**, where robots from different modules work together on one farm — sowing, scouting, weeding, climbing, cleaning and carrying — and hand work to each other.

Design → Build → Simulate → Swap → Deploy → Collaborate

2. THE SEVEN KRISHI MODULES

Module	Name	Indian problem it solves	Kit
K1	Drishti — Plant Health	Late detection of disease, pests, nutrient and water stress	Kit 1
K2	Vahan — Field Transport	Workers carrying heavy loads over long, uneven farm paths	Kit 1
K3	Nirai — Weeding	Weeding is labour-heavy; herbicide overuse harms soil and health	Kit 1 + module, or Kit 2
K4	Buvai — Seeding	Uneven seed spacing and depth reduce yield and waste seed	Kit 1 + module, or Kit 2
K5	Adike Aaro hi — Arecanut Climber	Dangerous climbing for spraying and harvesting; scarce skilled climbers	Kit 2
K6	Kalpa Aaro hi — Coconut Climber	Shortage of coconut climbers; risk of falls; delayed harvest	Kit 2
K7	Jal Mitra — River-Weed Cleaner	Water hyacinth and floating weeds choking lakes, rivers and canals	Kit 2

Teams may enter one module or many. Module winners are crowned for each module, and the overall **Sapta Krishi Champion** is the team that best demonstrates breadth, modularity and integration.

3. CHALLENGE VISION

Building India's modular agri-robotics ecosystem. The challenge aims to prove that affordable, reusable robot building blocks can be reconfigured for many farm jobs — the way one tractor carries many implements.

Intelligence	Mechatronics	Modularity & Impact
• Computer vision on the edge • Plant-health AI • Crop vs weed recognition • SLAM & navigation • Decision making	• Climbing mechanisms • Floating platforms • Weeding & seeding tools • Grippers & cutters • Waterproofing	• Plug-and-play tools • Shared software stack • Low-cost designs • Farmer usability • Multi-robot teamwork

Sense → Understand → Decide → Act → Swap → Collaborate

4. CHALLENGE OBJECTIVE

To design modular agricultural robots, using the ARJUNA AMR and the Jetrick controller, that autonomously perform real Indian farm tasks safely, reliably and affordably — and that can share tools, data and work with one another.

Across the seven modules, the challenge tests:

1. Perception: plants, weeds, seeds, fruit, nuts, trunks and floating debris
2. Navigation on farm paths, plantations and water
3. Mechanism design: climbing, cutting, pulling, sowing, collecting
4. Control: grip, balance, buoyancy, force and position
5. Autonomy and decision making under uncertainty
6. Failure detection and recovery
7. Modularity: quick tool swap and automatic recognition
8. Multi-robot coordination and hand-offs
9. Safety around people, heights, blades and water
10. Cost and practicality for Indian farmers

5. THE TWO KITS

Kit 1 — ARJUNA Autonomous Mobile Robot

Component	Specification	Role in the challenge
Platform	Fully assembled autonomous mobile robot	Same hardware for every team — a fair software contest
2D LiDAR	Planar laser scanner	Mapping, localisation, obstacle avoidance
OAK-D 3D stereo depth camera	Stereo depth + colour camera with on-device AI processing	Plant health, weed & seed detection, 3D measurement
Compute	NVIDIA Jetson Orin Nano Super, 8 GB	Edge AI, ROS 2, navigation, module management
Software	ROS 2-based	Common interfaces for all modules

Component	Specification	Role in the challenge
Expansion	Top-plate mount via KMI (specification TBC)	Carries tool modules for K2, K3, K4

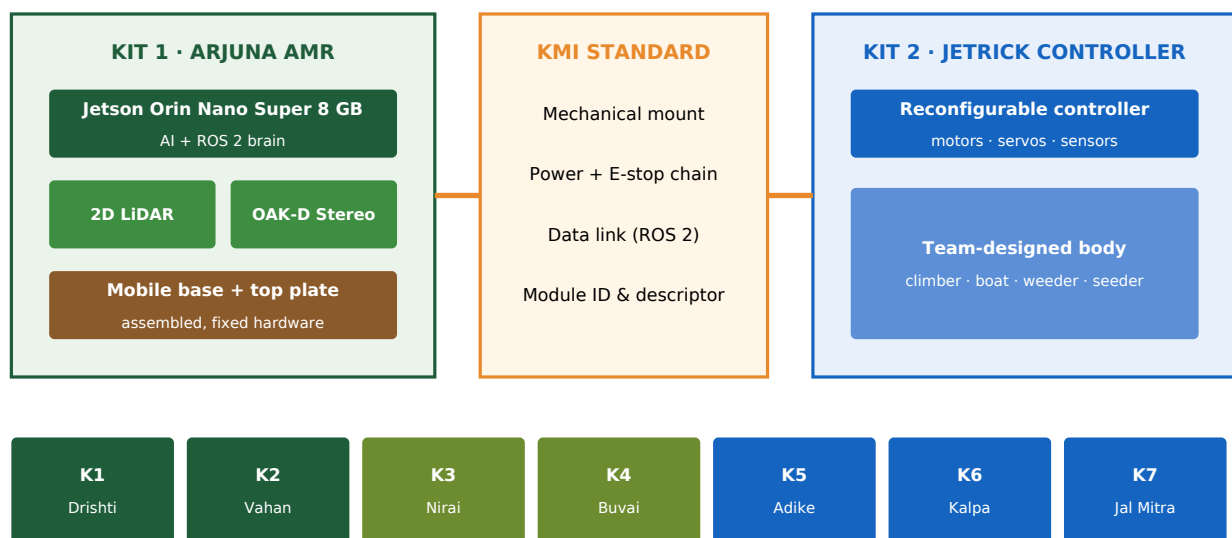
Kit 2 — Jetrick Robot Controller

Feature	Description
Reconfigurable control	Drives team-chosen motors, servos and actuators; reads team-chosen sensors
Design freedom	Teams build their own robot body to suit the task — climber, boat, weeder, seeder
KMI connection	Links to ARJUNA as a 'tool module', or runs a standalone robot
Detailed specification	I/O count, motor channels, supported sensors and programming tools to be published in the Rulebook

Build budget. To keep the contest fair and farmer-relevant, Kit 2 teams may add mechanical parts, motors, sensors and batteries up to a published cost cap (amount TBC). Every team declares a bill of materials.

6. THE MODULAR SYSTEM — KRISHI MODULE INTERFACE (KMI)

ONE BRAIN, MANY BODIES — the Krishi Module Interface (KMI)



Green = Kit 1 · Olive = Kit 1 + module or Kit 2 build · Blue = Kit 2 build

The KMI is the heart of the challenge. Every module built with Kit 2 that mounts on ARJUNA must follow it:

KMI layer	Requirement
Mechanical	Standard mounting pattern and quick-release fixing on ARJUNA's top plate / rear hitch
Power	Defined voltage and current limits; module draws power safely or carries its own battery

KMI layer	Requirement
Safety	Every module joins the robot's emergency-stop chain — pressing E-stop stops the tool too
Data	ROS 2 link between Orin and Jetrick with standard topics (start, stop, status, tool data)
Identity	Each module announces a Module Descriptor: ID, type, capabilities, limits
Auto-configure	ARJUNA detects the module and loads the matching behaviour automatically

A farmer should be able to change ARJUNA from a transport robot into a weeding robot in minutes, without a laptop.

7. COMPETITION LEAGUES

League 1 — ARJUNA League	League 2 — Jetrick Build League	Grand League — Sapta Krishi
Kit 1 · fixed hardware • K1 Drishti and K2 Vahan • Software, AI and autonomy focus • Optional K3 / K4 using a KMI tool <i>For AI, CV and software teams</i>	Kit 2 · design your own • Any of K3-K7 • Mechanism + control + autonomy • Cost and robustness matter <i>For mechanical & mechatronics teams</i>	Both kits · integrated • At least 3 modules, using both kits • KMI Pit Stop swap challenge • Krishi Mela Farm Day finale <i>The Grand Challenge</i>

Autonomy multiplier. Autonomy is the goal, but some modules are very hard. To keep participation open, scores are multiplied by the autonomy level achieved in each run:

Autonomy level	Description	Multiplier
Full autonomy	Human only starts the run and handles safety	× 1.0
Supervised autonomy	Human may confirm a key decision (e.g., 'cut this bunch?') — logged	× 0.8
Assisted tele-operation	Human drives; robot handles low-level control (Bronze tier only)	× 0.5

8. MODULE K1 — DRISHTI: COMPLETE PLANT HEALTH DETECTION

K1 · DRISHTI — Complete Plant Health

Kit 1 · ARJUNA League

Using only its on-board camera, ARJUNA patrols crop beds and produces a complete health card for every plant — not just 'healthy / diseased', but leaf, stem, fruit, growth and stress — so the farmer knows exactly what to do.

Arena

Raised beds or pots with around 40 plants (e.g., tomato, chilli, brinjal, beans) in mixed conditions, prepared and verified by agronomists. Plants are identified by bed and position.

The robot must report, for every plant

Health dimension	Examples of what to detect
Leaf condition	Leaf spots, blight, curl, mosaic, powdery coating, holes from chewing pests
Nutrient status	Yellowing patterns suggesting nitrogen, potassium or other deficiencies
Water status	Wilting, drooping, dry soil vs. waterlogging
Pests	Visible insects, eggs, webbing, sticky residue
Growth	Plant height and canopy size from depth data; growth stage
Fruit / flower	Flower count, fruit count, damaged or diseased fruit
Overall score	A simple 0-10 health score with the top recommended action

Difficulty tiers

Bronze	Silver	Gold
Single crop, clear symptoms, labels on pots	Mixed crops, early-stage symptoms, some occlusion	Unlabelled positions, look-alike symptoms (dust, water marks, natural variegation), changing light, plants moved between runs

Scoring (100)

Criterion	Points
Problem detection accuracy (recall & precision vs. agronomist truth)	35
Correct plant / position assignment	10
Growth measurement accuracy (height, canopy)	10
Coverage & navigation (all plants, no collisions)	15
Robustness to twists	10
Health-card clarity & farmer usefulness (incl. regional language)	10
Edge efficiency (runs fully on the Orin, no cloud)	10

9. MODULE K2 — VAHAN: FIELD TRANSPORT OF GOODS

K2 · VAHAN — Farm Goods Transport

Kit 1 · ARJUNA League

Farm workers spend hours carrying fertiliser bags, seed trays, tools and harvest crates. ARJUNA must act as a smart farm carrier — taking delivery orders, following workers on request and moving goods safely over farm paths.

Arena

A farm track with a gentle slope, speed bumps, a loose-soil patch, a narrow bund-like path and several loading points (store, field, pack-house). Payloads stay within ARJUNA's rated capacity (to be published).

Tasks

- **Delivery orders:** 'Take 2 seed trays from the store to Bed 4', received as a list during the run
- **Follow-me mode:** follow a specific worker in a group, ignore others, stop when the worker stops
- **Milk-run:** plan the shortest route to serve multiple orders with limited capacity
- **The Lassi Test:** carry an open glass of water on top across the track — spill as little as possible (smooth control)
- **Load check:** use the camera to confirm the right item was loaded before leaving

Twists

A gate closes, a cow-sized obstacle appears, a worker changes the destination mid-route, a path becomes muddy, a second worker wearing similar clothes walks beside the one being followed.

Scoring (100)

Criterion	Points
Orders delivered correctly and on time	30
Route efficiency & capacity planning	15
Follow-me accuracy (right person, safe distance)	15
Smoothness — Lassi Test spill score	10
Load verification by camera	10
Safety around people and obstacles	10
Robustness & recovery	10

10. MODULE K3 — NIRAI: WEEDING MECHANISM

K3 · NIRAI — Mechanical Weeding

Kit 1 + KMI tool, or Kit 2 build

Weeding is one of the most labour-intensive jobs in Indian farming, and herbicide overuse harms soil and people. The robot must find weeds among young crops and remove them mechanically — without harming the crop.

Arena

Soil beds with crop seedlings in rows (e.g., onion, maize or vegetable seedlings) and weeds both between rows (inter-row) and close to crop plants (intra-row — the hardest case). Real or realistic artificial weeds with measurable pull-out force are used.

Rules

- Mechanical / physical removal only — no chemical sprays.
- Crop plants damaged or uprooted are heavily penalised.
- Tool may be a rotating tine, pull-gripper, cutter, flame-free thermal or other method approved for safety.
- Blades must be guarded and must stop instantly with the E-stop.

Difficulty

Gold tier adds weeds that look like the crop (e.g., grass weeds among maize), weeds touching crop stems, uneven rows and wet soil.

Scoring (100)

Criterion	Points
Weeds removed (count and root removal)	30
Crop safety (damage penalties)	25
Weed vs crop recognition accuracy	15
Coverage and speed (area per minute)	10
Mechanism design, durability & cost	10
Modularity (KMI fit, quick mount)	10

11. MODULE K4 — BUVAI: PRECISION SEEDING SYSTEM

K4 · BUVAI — Precision Seeding

Kit 1 + KMI tool, or Kit 2 build

Hand sowing gives uneven spacing and depth, wasting seed and lowering yield. The robot must sow seeds at the right spacing, depth and count along straight rows — and record exactly where each seed went.

Arena

A prepared soil bed with marked boundaries. Seed type announced in the Rulebook (larger seeds such as maize, groundnut or beans for the first edition). Hidden stones and an existing plant patch must be avoided.

Tasks

- Sow N rows at the specified row spacing and plant spacing
- Place each seed at the target depth and cover it
- Exactly one seed per spot (singulation) — doubles and misses are counted
- Skip obstacles (stone, existing plant) and mark them on the map
- Publish a seed map: position of every seed sown
- **Ankur bonus:** the bed is watered and germination is counted after about 7-10 days — real results earn bonus points

Scoring (100)

Criterion	Points
Spacing accuracy (row and plant)	25
Depth accuracy (checked by organisers at random points)	20
Singulation (no doubles, no misses)	15
Seed-map accuracy	10
Obstacle handling	10
Mechanism design, cost & modularity	20
Ankur germination bonus	+5

12. MODULE K5 — ADIKE AAROHI: ARECANUT TREE CLIMBING ROBOT

K5 · ADIKE AAROHI — Arecanut Climber

Kit 2 · Jetrick Build League

In arecanut plantations of Karnataka, Kerala and the North-East, workers climb tall, slender palms to spray against fruit rot during the monsoon and to harvest bunches. It is skilled, dangerous work, and climbers are increasingly hard to find. The robot must climb, inspect, spray and harvest — safely.

Arena

Practice and early rounds use test trunks (real arecanut logs or bark-textured poles) of slender diameter, fixed vertically to a safe height (e.g., 6–8 m) with fall-arrest netting. Dummy nut bunches with different maturity colours are fixed near the top. The Grand Finale may use real palms at a partner plantation under strict safety control.

Tasks

- Mount on the trunk (a human may place the robot at the base) and climb to the crown
- Detect and count bunches; classify ripe (yellow-orange) vs. unripe (green)
- **Spray mission:** spray water (stand-in for fungicide) onto the target bunches only — measured by water-sensitive cards
- **Harvest mission:** cut the stalk of the ripe bunch only and lower it on a rope or in a basket — no free fall
- Descend in a controlled way and report what it saw

Twists

Trunk diameter changes along the height, a wet / slippery section (monsoon), a bulge or scar on the trunk, wind from a fan, a 'no-climb' zone marked mid-trunk, and in Gold tier, a second bunch that looks ripe but is not.

Scoring (100)

Criterion	Points
Climb success and height reached	20
Grip stability & controlled descent (no slip, no trunk damage)	15
Bunch detection & ripeness classification	15
Spray accuracy (on target, minimal drift)	15
Harvest: correct bunch cut and safely lowered	20
Weight, set-up time, cost & farmer practicality	15

13. MODULE K6 — KALPA AAROHI: COCONUT TREE CLIMBING ROBOT

K6 · KALPA AAROHI — Coconut Climber

Kit 2 · Jetricks Build League

The coconut palm — the Kalpavriksha — supports millions of farmers, but trained climbers are scarce and the work is risky. The robot must climb thicker, often curved trunks, choose the right nuts and bring them down without cracking them.

Arena

Thicker test trunks with leaf-scar rings, including one inclined trunk, fixed to a safe test height with fall-arrest netting. A simulated crown with fronds, and bunches of dummy nuts at tender (green) and mature (brown) stages. Real palms may be used in the finale under strict safety control.

Tasks

- Climb past frond bases into the crown without getting stuck
- **Order-based harvest:** 'Bring 3 tender coconuts' or 'Bring 1 mature bunch' — choose correctly
- Cut the selected bunch / nuts and lower them in a net or basket — dropped nuts are penalised
- **Crown health check:** photograph the crown and report signs such as damaged fronds, holes or pest marks
- Descend and hand the harvest to the K2 transport robot (Farm Day)

Twists

Inclined trunk, a dry frond blocking the path, nuts on the far side of the crown, a mixed bunch of tender and mature nuts, a sudden gust.

Scoring (100)

Criterion	Points
Climb success on vertical and inclined trunks	20
Crown navigation (passing fronds safely)	10
Correct nut selection for the order	20
Safe cutting & lowering (no dropped or cracked nuts)	20
Crown health report	10
Weight, set-up time, cost & practicality	20

14. MODULE K7 — JAL MITRA: RIVER-WEED CLEANING ROBOT

K7 · JAL MITRA — Floating Weed Cleaner

Kit 2 · Jetrick Build League

Water hyacinth and other floating weeds spread fast across Indian lakes, rivers and irrigation canals — blocking water flow, harming fish and breeding mosquitoes. The robot must float, find the weed, collect it and bring it to shore, while separating plastic waste.

Arena

A test pool or canal section with floating artificial weed mats (realistic size, buoyancy and drag), floating plastic bottles and wrappers, buoys marking a no-go zone, and pumps or fans creating current and wind. The finale may move to a partner lake or canal with environmental permission.

Tasks

- Launch from the shore station and map the weed patches
- Collect weed into an on-board store (conveyor, rake, net or team's own design)
- Separate plastic from weed into a different compartment
- Return to shore and unload into the correct bins before the store overflows
- Avoid buoys, ducks/fish decoys and the no-go zone; never cut through 'protected' water plants marked by organisers
- Report: area cleared, weight collected, plastic count, remaining weed map

Twists

Current reverses, a weed mat tangles the propeller zone, a floating log blocks the route, low battery, sun glare on water.

Scoring (100)

Criterion	Points
Weed collected (weight / area cleared)	30
Plastic separation accuracy	10
Navigation on water (no-go zones, obstacles, current)	15
Shore unloading & cycle efficiency	10
Waterproofing, stability & recovery (anti-tangle, self-righting)	15
Environmental care (no harm to protected plants / decoys)	10
Cost & practicality for panchayats / water bodies	10

15. THE KMI PIT STOP — MODULE SWAP CHALLENGE

Part of the Grand League. ARJUNA arrives at the Pit Stop carrying the transport tray (K2). The team must:

1. Remove the transport tray and fit their weeding (K3) or seeding (K4) module
2. Let ARJUNA detect the new module through the KMI and load the right behaviour automatically
3. Pass a safety check (E-stop stops the tool) and start the new task

Measure	Points
Swap time (human hands-on time)	10
Automatic recognition (no laptop, no code change)	10
Safety-chain check passed	5
Task started correctly after swap	5

Fastest swap with perfect safety wins the **Pit Stop Award**.

16. GRAND FINALE — KRISHI MELA FARM DAY

The finale brings modules together on one farm, in one integrated mission with hand-offs between robots.

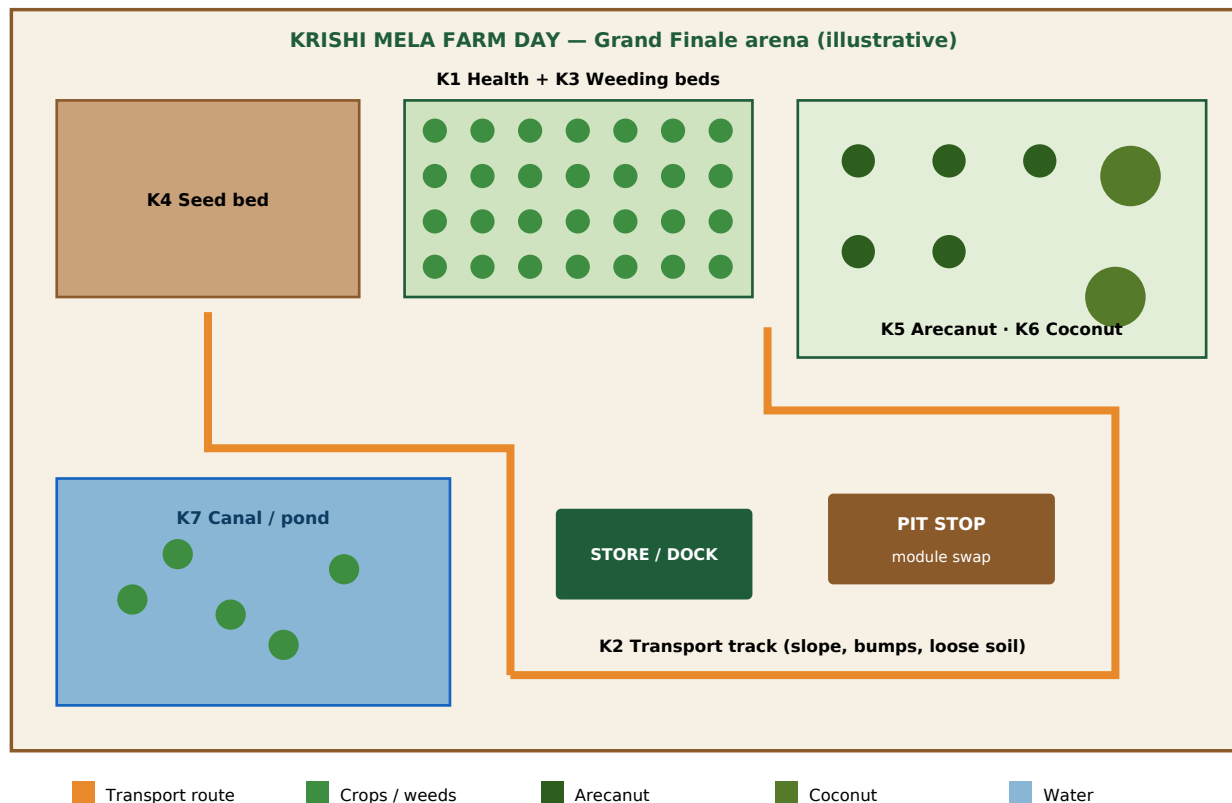


Figure 2 — Illustrative Farm Day arena with zones for all seven modules.

Time slot	Activity	Hand-off
Morning	K4 sows the seed bed; K2 brings seed trays from the store	K2 → K4
Mid-morning	K1 patrols crop beds and flags weedy and unhealthy plants	K1 map → K3
Late morning	Pit Stop: ARJUNA swaps to K3 and weeds the beds flagged by K1	KMI swap
Noon	K5 / K6 climb palms and harvest; harvest lowered into a crate	K5/K6 → K2
Afternoon	K2 collects the plantation harvest and delivers it to the pack-house	K2 → store
All day	K7 clears the irrigation canal so water can reach the beds	K7 → farm report
Surprise	A 'Mela Mayhem' event (rain shower, blocked path, robot breakdown) forces re-planning	All

A team in the Grand League fields its own robots for at least three modules. Modules the team has not built may be filled by the organisers' reference robots, so every team runs a complete Farm Day.

17. OVERALL SCORING — SAPTA KRISHI CHAMPION

Component	Points
Best three module scores (each scaled to 100, averaged)	50
Krishi Mela Farm Day performance	25
KMI Pit Stop	10
Modularity & reuse (shared parts, software and design across modules)	10
Affordability & farmer impact (cost, ease of use, repairability)	5
TOTAL	100

18. SIMULATION, DESIGN REVIEW & DIGITAL TWINS

Kit	Pre-hardware stage
Kit 1 (ARJUNA)	ARJUNA digital twin with LiDAR and OAK-D models in Gazebo / Isaac Sim; virtual farm beds and transport track; containerised ROS 2 submissions scored on hidden scenarios
Kit 2 (Jetricks)	Design Review: CAD model, mechanism calculations (grip force, buoyancy, torque), safety analysis, bill of materials and a simulation or small-scale prototype video

Training and competition environments will differ: teams train on public scenarios and are scored on hidden ones, as in established autonomous-systems competitions.

19. COMPETITION STAGES

Stage 0 — Launch

9-10 October 2026 (proposed) — Challenge reveal, ARJUNA and Jetrick demonstrations, module showcase, registration opens.

Stage 1 — Registration & Module Selection

October – November 2026 — Teams choose their league and modules.

Stage 2 — Onboarding Bootcamps

November 2026 — ROS 2, Orin & OAK-D, Jetrick, KMI, mechanism design, farm safety; plantation and lake field visits.

Stage 3 — Simulation & Design Review

November – December 2026 — Kit 1 teams submit to the simulator; Kit 2 teams submit designs.

Stage 4 — Kit Allocation

December 2026 — Shortlisted teams receive kits on loan or booked lab access at NMIT.

Stage 5 — Build & Test

December 2026 – February 2027 — Hardware builds, practice arenas, mentor sessions, safety inspections.

Stage 6 — Module Trials (Bronze / Silver)

February 2027 — Each module's arena opens for scored qualification runs.

Stage 7 — Module Finals (Gold)

March 2027 — Module champions decided; Grand League teams selected.

Stage 8 — Krishi Mela Farm Day

April 2027 — Pit Stop, integrated Farm Day and awards.

20. SAFETY

Safety is mandatory and is checked before any robot runs.

Module	Key safety rules
All	Hardware E-stop and remote stop; every KMI module on the E-stop chain; safety inspection before each run
K2 Transport	Speed limits near people; payload limits; secure loads
K3 Weeding / K4 Seeding	Guarded blades and moving parts; tool stops instantly on E-stop
K5 / K6 Climbing	Test-height limit, fall-arrest net or safety tether on every robot, exclusion zone under the tree, helmets for all crew, controlled lowering of all cut material
K7 Water	Retrieval line, rescue boat or pole, life jackets for crew, waterproof batteries, no fuel engines

A severe safety violation may result in disqualification irrespective of the numerical score.

21. HUMAN INTERVENTION

Allowed	Not allowed (in full-autonomy runs)
✓ Starting the run ✓ Placing a climber at the base of the trunk ✓ Launching the boat from the shore station ✓ Emergency / safety stops ✓ Module swap at the Pit Stop (timed)	✗ Steering the robot ✗ Choosing targets (weeds, nuts, bunches) for the robot ✗ Pushing or lifting the robot during a task ✗ Moving arena items to help the robot ✗ Correcting reports by hand

Runs with supervised autonomy or tele-operation are scored with the autonomy multiplier (Section 7).

22. PARTICIPATION & ELIGIBILITY

Category	Description
Student Team	Undergraduate / postgraduate students
Research Team	University and research-institute teams, including agricultural universities
Startup Team	Agri-tech and robotics startups
Industry Team	Corporate R&D; teams
Open / Farmer-Innovator Team	Independent makers and farmer-innovators

Team size: 3-8 members recommended (Kit 2 builds need mechanical, electronics and software skills). Teams entering climbing or water modules must include a member responsible for safety. Physical stages are subject to venue, safety and legal requirements; international teams may join simulation and design stages.

23. SUBMISSION REQUIREMENTS

Item	Details
Software	Containerised ROS 2 package (Kit 1) / controller code and ROS 2 nodes (Kit 2)
Design dossier (Kit 2)	CAD, mechanism calculations, safety analysis, bill of materials with costs
KMI declaration	Module Descriptor and interface compliance checklist
Technical report	5-10 pages per module
Demo video	3-5 minutes per module
Farmer pitch (finalists)	2-minute plain-language explanation of cost, use and value
Team declaration	Originality, licences, safety compliance

24. OPEN SOURCE, DATA & INTELLECTUAL PROPERTY

- Background IP stays with the team; challenge-developed IP stays with participants unless otherwise agreed.
- The KMI specification itself will be published openly so anyone can build compatible modules.
- Teams are encouraged to open-source designs and code; the best open contribution wins the Open Krishi Award.
- With consent, plant-health and weed images may form an open Indian agri-robotics dataset.
- Kit hardware, digital twins and evaluation tools are governed by the Challenge Terms of Use.

25. PARTNER ROLES & VALUE

Partner	Contribution	Value
NMIT (Organiser)	Organisation, arenas, labs, evaluation, outreach, Farm Day	National leadership in modular agri-robotics
ARJUNA platform partner	Kit 1 robots, digital twin, technical support	Large developer community; new agri applications
Jetrick partner	Kit 2 controllers, documentation, mentoring	Dozens of new robot designs built on its controller
Agricultural university	Crops, ground truth, agronomy guidance, jury	Applied research and student talent
Plantation / FPO partners	Arecanut & coconut sites, farmer panels	Early access to climbing and harvesting robots
Agri-tech industry	Problem validation, prizes, pilots	Pilot-ready solutions and recruitment

26. GOVERNANCE & JURY

Committee	Responsible for
Steering Committee	Direction, partnerships, budget, major decisions
Technical & KMI Committee	Kits, KMI standard, digital twins, arenas, technical rules
Agronomy Committee	Crops, ground truth, plantation & water-body coordination
Safety Committee	Height, water, blade and robot safety; inspections
Independent Evaluation Committee	Scoring, verification, disputes

The jury may include experts from NMIT, platform partners, agricultural universities, plantation-crop research institutes, IITs / IISc, agri-tech startups, FPOs and practising farmers and plantation workers.

27. PROPOSED AWARDS

Award	Criteria
Sapta Krishi Champion	Highest overall Grand League score
Module Champions (×7)	Best team in each of K1-K7
Pit Stop Award	Fastest, safest KMI module swap
Aarohi Award	Best climbing mechanism (K5 / K6)
Jal Rakshak Award	Most effective and eco-friendly water robot
Sahyog Award	Best multi-robot hand-offs on Farm Day
Kisan Jugaad Award	Best performance-per-rupee design
Open Krishi Award	Best open-source design or dataset
Best Student Innovation	Outstanding student-led team
Best Startup Pathway	Strongest route to a real product

Prize amounts (₹X,XX,XXX per tier) and non-cash benefits — kits for winning institutions, plantation pilots, incubation, internships and mentorship — will be announced separately.

28. PROPOSED TIMELINE

Phase	Proposed timeline
Launch & registration opens	9-10 Oct 2026
Registration & module selection	Oct - Nov 2026
Bootcamps & field visits	Nov 2026
Simulation & design review	Nov - Dec 2026
Kit allocation	Dec 2026
Build & test	Dec 2026 - Feb 2027
Module trials (Bronze / Silver)	Feb 2027
Module finals (Gold)	Mar 2027

Phase	Proposed timeline
Krishi Mela Farm Day & awards	Apr 2027

Final dates will be published on the Challenge Portal. Plantation and water-body rounds will be scheduled with partners to suit weather and crop conditions.

29. WHAT CONSTITUTES A SUCCESSFUL SOLUTION?



30. LONG-TERM VISION

Future edition	Direction
More modules	Spraying drones docking on ARJUNA, fruit harvesting arms, cattle-shed cleaning, grain drying
Fleets	Many ARJUNA and Jetrick robots sharing one farm map and task list
KMI ecosystem	Third-party companies selling KMI-compatible tools to farmers
Field deployment	Pilot programmes with FPOs, plantations and panchayats

Seven modules → One platform → A farmer-owned robot toolkit for Bharat

CONTACT & DOCUMENT STATUS

Field	Value
Secretariat	Sapta Krishi Modular Challenge Secretariat
Institution	Nitte Meenakshi Institute of Technology, Nitte (Deemed to be University), Bengaluru
Department	Department of Robotics & Artificial Intelligence · Centre for Robotics Research
Email / Website	[to be announced]
Version / Status	1.0 — Draft for Partner Review

Kit specifications, the KMI standard, arena dimensions, scoring values, safety procedures, prize structure and IP terms will be published in the official Technical Rulebook and Terms & Conditions before the competition begins.